

# CSIT305 — Study Guide & Mock Exam

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*Based on Lecture 7 (Subject Revision) topic outline*

## Exam facts to keep in mind

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- **Format:** 8 essay questions —  $6 \times 5$  marks (short answer) +  $2 \times 10$  marks (long answer) = 50 marks total
  - **Weight:** 50% of final grade. Need  $\geq 40\%$  on the exam (20/50) to avoid Technical Fail.
  - **Closed book, 3 hours.**
  - **Hint from lecturer:** test understanding, not memorisation. Quality over quantity.
  - Recurring exam patterns (from both sample questions given):
    1. **"Technique X degrades under condition Y — explain why, propose a fix"** (structural/algorithmic reasoning)
    2. **"Given this real-world scenario, apply multiple data/query types and explain how they help"** (applied, multi-topic synthesis)
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## Topic Checklist (tick off as you review)

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| # | Lecture                    | Core Concepts                                                                                                                                   |
|---|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Advanced Database Overview | Structured/unstructured & big data, relational vs non-relational, hierarchical, network, NoSQL, graph, document DB, streaming systems, cloud DB |

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|----|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2  | Spatial Data and Computing     | Window/range/kNN queries, Quad-tree, R-tree, best-first search, IR-tree, spatial-textual queries                                                                  |
| 3  | Graph Data and Computing       | Nodes/edges, directed/weighted/bipartite/multigraphs, adjacency matrix/list, degree distribution, network diameter, Linear Threshold & Independent Cascade models |
| 4  | Text Data and Processing       | Text mining, classification vs clustering, NLP tasks (POS tagging, parsing, NER, WSD, semantic role labeling)                                                     |
| 5  | AI4SE                          | Software effort estimation, requirement elicitation, code completion / API recommendation via software mining                                                     |
| 6  | SE in Cloud Computing I & II   | Cloud service/deployment models, monolithic vs microservices, cloud requirement engineering                                                                       |
| 7  | IoT and Edge Computing         | IoT components (edge, communication, cloud), IoT design process, edge computing rationale                                                                         |
| 8  | Digital Twin Technology        | DT model (data collection, model training, application), DTP/DTI/DTA types                                                                                        |
| 9  | Blockchain and Smart Contracts | Blockchain basics/challenges, use categories, smart contracts, writing/using them                                                                                 |
| 10 | Emerging Cyber-Attacks         | Attack trends, ransomware, IoT botnets, cryptojacking, defense systems                                                                                            |

## Mock Short-Answer Questions (5 marks each)

### Databases

1. Compare relational and non-relational (NoSQL) databases in terms of schema flexibility, scalability, and typical use cases.
2. Explain the structural difference between a hierarchical and a network database model, including one advantage and one disadvantage of each.

**Spatial Data** 3. Explain how a k-nearest-neighbour (kNN) spatial query differs from a range query, and describe how best-first search is used to process one of them efficiently. 4. What is an IR-tree, and how does it extend a spatial index to support spatial-textual queries?

**Graph Data** 5. Define network diameter and degree distribution, and explain what each tells you about the structure of a real-world network (e.g., a social network). 6. Compare the Linear Threshold Model and the Independent Cascade Model for influence maximisation — what is the key difference in how influence spreads?

**Text Data** 7. Explain the difference between text classification and text clustering, including one example use case for each. 8. Choose two NLP core technologies (e.g., Named-Entity Recognition, Part-of-Speech tagging) and explain what problem each solves.

**AI4SE** 9. Explain how AI methods can be applied to software effort estimation, and why this is a difficult problem to automate. 10. What is API code recommendation, and how does software mining support it?

**Cloud Software Engineering** 11. Define the four cloud deployment models (public, private, hybrid, community) and explain one key requirement driver specific to cloud applications. 12. Explain two challenges of monolithic cloud applications that microservices architecture is designed to solve.

**IoT & Edge** 13. Explain the three major functional components of an IoT system and why edge computing is often preferred over pure cloud processing for IoT data.

**Digital Twin** 14. Explain the difference between a Digital Twin Prototype (DTP), Digital Twin Instance (DTI), and Digital Twin Aggregate (DTA).

**Blockchain** 15. What problem do smart contracts solve that plain blockchain transactions cannot, and give one application example.

**Cybersecurity** 16. Explain how a botnet attack using IoT devices works, and describe one defense strategy against it. 17. What is cryptojacking, and how does it differ from ransomware in terms of attacker goals?

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## Mock Long-Answer Questions (10 marks each)

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### A. Technique failure / mitigation pattern (matches your given sample)

1. Both Quadtrees and R-trees may experience performance degradation under certain data distributions. Explain the underlying reasons for this issue and propose appropriate mitigation strategies for each case. (*given sample — see my earlier answer*)
2. Graph databases and relational databases can both experience performance issues at scale, but for different structural reasons. Explain why deeply interconnected data causes problems for relational databases (e.g., excessive JOINS), and why graph databases avoid this — including one scenario where a graph database itself would still struggle.
3. NLP systems built on core techniques like POS tagging, parsing, and NER can degrade badly on informal or domain-specific text (e.g., social media, medical text). Explain why, and propose mitigation strategies (e.g., domain adaptation, fine-tuning).

**B. Applied/scenario synthesis pattern** (matches your given sample) 4. (*given sample*) You work for a transportation company managing a fleet of vehicles, tracking routes, schedules, and maintenance histories. Explain how a multi-modal database could store and manage this data, referencing at least two data types, at least three relevant queries, and the processing technique for each. 5. A hospital wants to build a system that stores patient records (structured), doctors' notes (text), MRI scan locations within the hospital (spatial), and referral relationships between doctors (graph). Design a multi-modal database approach for this scenario, explaining which database type suits each data type and one query example per type. 6. A logistics company wants to deploy an IoT

and Digital Twin solution to monitor its delivery trucks in real time. Explain the roles of edge computing, cloud computing, and Digital Twin modelling in this system, and discuss one cybersecurity risk this architecture introduces along with a mitigation. 7. A retail company is deciding between a monolithic and a microservices architecture for its new cloud-based e-commerce platform, which needs to scale rapidly during sales events. Discuss the cloud application requirements at play, and justify which architecture is more appropriate, referencing at least two specific challenges.

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## How to use this guide

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1. Go topic by topic through the checklist — for each, write your own 3-4 sentence definition **without looking at slides**, then check yourself.
2. Attempt the 5-mark questions timed (~4 minutes each) — aim for 2-3 precise points, not a full essay.
3. Attempt the 10-mark questions timed (~12-15 minutes each) — these expect structure: a clear framework/definition, application to the scenario, and justification.
4. Redo any question where you got stuck 2-3 days later (spaced repetition) rather than moving straight to the next topic.